IoT SYSTEM FOR SEWAGE WORKER SAFETY

Minor project-1 report submitted
in partial fulfillment of the requirement for award of the degree of

Bachelor of Technology
in
Computer Science and Design

By

MANNE BANDHAVI	(22UECE0037)	**(VTU 21991)**
KANCHARLA SUMANTH	(22UECE0028)	**(VTU 23499)**
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Under the guidance of
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DEPARTMENT OF COMPUTER SCIENCE AND DESIGN
SCHOOL OF COMPUTING

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY

(Deemed to be University Estd u/s 3 of UGC Act, 1956)
Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA

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CERTIFICATE

It is certified that the work contained in the project report titled “IoT SYSTEM FOR SEWAGE WORKER SAFETY” by “MANNE BANDHAVI (22UECE0037), MORUGU PAVAN SAI (22UECE0 039), KANCHARLA SUMANTH (22UCECE0028)” has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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November, 2024

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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APPROVAL SHEET

This project report entitled IoT SYSTEM FOR SEWAGE WORKER SAFETY by MANNE BANDHAVI (22UECE00037), M.PAVAN SAI (22UECE0039), K.SUMANTH (22UCECE0028) is approved for the degree of B.Tech in Computer Science and Design .

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Place:

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ABSTRACT

In modern urban environments, ensuring the safety of sewage workers is critical due to the hazardous conditions they encounter. This project presents an Internet of Things (IoT) system designed to enhance the safety of sewage workers by integrating various sensors with an ESP8266 WiFi module for real-time monitoring and alerting. The system employs an ultrasonic sensor to measure the depth of sewage, a gas sensor to detect harmful gases, and a DHT11 sensor to monitor temperature and humidity levels within the sewage system. Data collected from these sensors are transmitted via the ESP8266 WiFi module to a cloud-based platform for continuous monitoring. Alerts and notifications are generated based on predefined safety thresholds, enabling timely interventions and enhancing worker safety. This IoT-based approach provides a comprehensive solution for real-time safety management in hazardous sewage environments, potentially reducing risks and improving overall operational safety.

Keywords:

IoT (Internet of Things), Sewage worker safety, ESP8266 WiFi module, API (Application Programming Interface), Real-time monitoring, Ultrasonic sensor, Gas sensor, DHT11 sensor.

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
DHT11	Digital Humidity and Temperature Sensor
ESP8266	A low-cost Wi-Fi microchip with microcontroller capability
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
RTOS	Real-Time Operating System
Wi-Fi	Wireless Fidelity

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Chapter 1

INTRODUCTION

1.1 Introduction

Sewage systems, while essential for maintaining urban health and sanitation, inherently create dangerous conditions for workers due to exposure to hazardous environments. These environments include toxic gases like methane and hydrogen sulfide, fluctuating temperatures, and rapidly changing sewage levels. Traditional safety measures, such as manual inspections, often fail to provide adequate early warnings of potential dangers. For instance, workers may not be aware of a sudden increase in sewage depth or rising gas concentrations, putting them at risk of accidents, respiratory issues, or environmental hazards. The absence of real-time monitoring and predictive alerts increases the chances of delayed responses to such dangers, further elevating the risk of injury and exposure to toxic elements.

To address these challenges, the proposed IoT-based safety system aims to continuously monitor sewage environments, offering real-time insights and timely alerts. The system uses an ultrasonic sensor to measure sewage depth, providing critical data about potential backups or overflows in the system. This sensor allows the system to detect abnormal water levels, which may indicate a blockage or potential flooding, thus giving workers early notice to take precautionary action before the situation worsens. This feature is especially important in avoiding hazardous exposure to floodwaters and ensuring that maintenance workers can avoid risky situations associated with sudden surges in sewage levels.

In addition to sewage depth monitoring, the system incorporates gas sensors capable of detecting harmful gases such as methane, ammonia, and hydrogen sulfide, which are commonly present in sewage systems. These gases can be hazardous to human health and, if undetected, may lead to severe respiratory problems, explosions, or poisoning. The gas sensors continuously analyze air quality, providing workers with real-time alerts if dangerous gas concentrations exceed safety thresholds. This early warning allows workers to evacuate the area or don protective gear before exposure becomes critical, greatly enhancing worker safety and minimizing the risk of accidents due to toxic gas inhalation.

The DHT11 sensor adds another layer of monitoring by tracking temperature and humidity levels within the sewage system. Extreme temperature variations and high humidity can further complicate safety, leading to conditions like heat stress, hypothermia, or even the growth of harmful bacteria. The system's ability to track these environmental parameters helps ensure that workers are not exposed to conditions that could adversely affect their health, such as working in excessively hot or cold conditions. All of this data is transmitted through an ESP8266 WiFi module to a cloud platform, where it is continuously monitored, processed, and analyzed. By providing real-time insights and sending instant alerts when safety parameters are breached, this IoT solution empowers workers and managers to respond proactively to risks, reducing injuries and improving overall hazard management in sewage systems.

1.2 Aim of the Project

The aim of this project is to design and implement an IoT-based safety monitoring system specifically for sewage workers. This system will leverage a combination of sensors and wireless communication technology to enhance worker safety by continuously monitoring and managing hazardous conditions within sewage environments.

1.3 Project Domain

The domain of this project lies in the intersection of public health, occupational safety, and Internet of Things (IoT) technology, focusing specifically on the safety management of sewage systems. Sewage systems are vital infrastructures in modern urban areas, ensuring the proper disposal and treatment of waste to maintain hygiene and prevent disease outbreaks. However, these environments are often hazardous for maintenance workers, who face risks from toxic gases like methane and hydrogen sulfide, temperature fluctuations, and unpredictable sewage levels.

Traditional safety practices, primarily relying on periodic manual checks, lack the responsiveness required for real-time hazard detection, leaving workers exposed to potential health risks and accidents. As urban infrastructure expands and the demand for efficient sanitation increases, the need for reliable, real-time safety solutions within sewage systems becomes even more critical.

1.4 Scope of the Project

The scope of this project encompasses the integration of multiple sensors and IoT technologies to establish a comprehensive safety monitoring system for sewage environments. The system will incorporate sensors that measure critical environmental factors, including gas levels, temperature, humidity, and sewage depth, each addressing a specific hazard that workers encounter in such settings. These sensors will be seamlessly connected to an ESP8266 WiFi module, enabling efficient data collection and wireless transmission to a cloud platform.

A key element of the project's scope is its use of cloud-based data analysis to detect and evaluate potential hazards. Real-time data will be continuously monitored and analyzed on the cloud platform to assess environmental conditions against established safety thresholds. When sensor readings indicate unsafe levels of gases, temperature, humidity, or sewage depth, the system will automatically generate alerts and notifications, ensuring immediate communication of these risks to relevant authorities. This automated alert mechanism not only enables proactive interventions but also minimizes response time in emergencies.

Chapter 2

LITERATURE REVIEW

[1] Subarna B., Kaviya Shri J.P., Roshika C., Sarulatha R., Dr. K. Suresh (2023): This project integrates multiple IoT-based sensors, such as MQ4 and MQ7, to detect methane and carbon monoxide in sewage systems. A microcontroller processes the data, and the system continuously monitors workers' health using wearable devices that track vital parameters. If gas levels exceed safe thresholds or health issues are detected, alerts are transmitted through GSM modules to emergency contacts and supervisors. This system aims to reduce accidents by providing early warnings and health monitoring to protect workers in real-time.

[2] Kanksha et al. (2022): The project focuses on a cloud-connected IoT sewage monitoring system. It employs Arduino Uno as the control unit, with gas sensors (MQ2 and MQ6) measuring harmful gases like hydrogen sulfide. The data is uploaded to ThinkSpeak, a cloud platform, where supervisors can monitor conditions remotely. If hazardous gas levels are detected, alerts are sent via GSM modules to relevant authorities. This system ensures that both environmental hazards and worker safety are continuously tracked, preventing exposure to dangerous gases during routine maintenance.

[3]Nanda Kumar et al. (2023): This work presents an IoT-enabled communication-based safety monitoring system that enhances real-time coordination between workers and supervisors. It uses gas sensors deployed within the sewer to monitor harmful gases like ammonia and hydrogen sulfide. The data is transmitted via Wi-Fi modules, and notifications are sent to emergency teams if gas levels cross the danger mark. The project highlights the importance of robust communication channels, enabling fast responses to emergency situations and enhancing the overall safety of workers.

[4] An Industrial IoT Approach (2023): This paper proposes a smart sewage gas identification and reporting system using intelligent IoT sensors and logical con-

trollers. The system monitors multiple gases, including methane, LPG, and carbon monoxide, to detect leaks and hazardous conditions in real-time. Data from these sensors is stored in the cloud, allowing municipal authorities to access it remotely. In addition to monitoring, the system provides predictive analytics, helping authorities take preventive actions before dangerous situations develop. This solution aims to automate routine checks, reduce risks, and improve decision-making through data analysis.

[5]Anna Ledin et al. (2023): This research investigates priority pollutants in sewage sludge, emphasizing the need for continuous monitoring to protect both workers and the environment. IoT-based sensors are used to detect a wide range of toxic chemicals, ensuring compliance with European environmental standards. The project identifies potentially harmful substances like pharmaceuticals and industrial chemicals that accumulate in sludge. By monitoring these pollutants, authorities can intervene early to protect workers and maintain environmental safety.

[6] B. Prakash and Team (2023): This project introduces wearable IoT technology for monitoring both environmental conditions and worker health. The wearable devices are equipped with gas sensors that detect harmful substances like hydrogen sulfide and methane. In addition, the system monitors heart rate and body temperature to ensure that workers remain healthy while performing maintenance tasks. Alerts are automatically sent to supervisors and healthcare facilities if unsafe conditions are detected, allowing rapid medical intervention.

[7] M. Aravind et al. (2022): This project emphasizes the use of smart helmets equipped with IoT-based sensors to enhance worker safety in hazardous environments. The helmets measure gas concentrations and monitor vital signs such as heart rate. If dangerous conditions are detected, such as high methane levels, the helmet sends alerts via Wi-Fi modules to remote monitoring centers. This system ensures that workers are protected not only from environmental hazards but also from health risks.

[8] G. Singh and Co-authors (2023): This project focuses on the development of a cloud-based IoT monitoring platform designed to track environmental conditions and worker health in real-time. The system uses multiple sensors to monitor temper-

ature, humidity, and toxic gases, while also tracking worker activity. Notifications are sent to emergency responders if dangerous situations arise, enabling fast action to prevent accidents. The project demonstrates the value of remote monitoring and cloud-based data analysis in enhancing safety protocols.

[9] R. Sharma et al. (2022): This work develops a multi-sensor platform to monitor gases such as ammonia, methane, and hydrogen sulfide. The system leverages predictive analytics to anticipate hazardous conditions, allowing authorities to implement preventive measures. The project highlights the importance of continuous monitoring in reducing risks associated with toxic gases and improving the safety of sewage workers through early intervention.

[10]P. Thomas et al. (2023): This project involves the design and implementation of a smart manhole monitoring system using IoT technology. The system detects blockages, water levels, and toxic gases in real-time, helping prevent accidents during maintenance operations. Alerts are sent automatically to supervisors and workers via mobile apps, ensuring quick response to hazardous conditions. The project aims to streamline sewer management and improve worker safety through automated monitoring.

The research papers collectively underscore recent IoT-based projects for sewage worker safety focus on monitoring hazardous gases like methane, ammonia, and carbon monoxide. These systems use sensors connected to microcontrollers to track environmental conditions, alerting supervisors through GSM modules or cloud platforms when gas levels exceed safe thresholds. Many solutions include wearable devices that monitor workers' vital signs, such as heart rate and temperature, to detect health risks in real-time.Some projects leverage smart helmets equipped with sensors, while others use cloud-based platforms for remote monitoring and predictive analytics like smart manhole monitors, detect blockages and toxic gases, reducing the need for manual inspections and minimizing exposure to dangerous environments. These IoT solutions enable faster emergency response, improving worker safety and reducing accidents during sewage maintenance.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

Existing sewage management systems rely heavily on manual inspections and periodic maintenance to monitor critical conditions within the sewage network. Workers are tasked with entering potentially hazardous environments to assess levels of harmful gases, sewage depth, and other environmental factors. While these inspections help prevent some issues, the inherent delay between checks creates a lag in hazard detection, making workers susceptible to sudden environmental changes. Additionally, manual checks are time-intensive and labor-intensive, leading to high operational costs and inefficiencies.

Another disadvantage of the existing system is its susceptibility to human error, which can lead to missed or incorrect readings of critical parameters like gas levels and sewage overflow risks. With urban populations on the rise, the sewage infrastructure is under greater pressure, increasing the likelihood of critical failures, such as blockages or gas buildups, which may go unnoticed until they become severe. This limited effectiveness and responsiveness underscore the need for a smarter, real-time monitoring approach to enhance worker safety, optimize maintenance efforts, and enable proactive decision-making.

3.2 Problem Statement

The proposed IoT-based sewage monitoring system aims to address the limitations of manual inspections by introducing real-time, automated hazard detection to improve safety and efficiency. By integrating sensors like ultrasonic, gas, and environmental sensors with an ESP8266 Wi-Fi module, this system enables continuous, remote monitoring of sewage conditions. Data from the sensors is transmitted to a cloud platform, where it is analyzed and visualized in real-time, allowing stakeholders to make informed decisions swiftly. Automated alerts based on predefined safety thresholds ensure that relevant authorities receive instant notifications of any hazardous conditions, enabling immediate intervention and reducing the risks to sewage workers.

This IoT solution offers significant advantages over traditional methods. It minimizes direct exposure of workers to dangerous conditions, improves data accuracy, and supports more efficient, data-driven infrastructure management. Furthermore, the modularity of the design allows for easy adaptation to other smart city applications, creating a versatile solution for urban safety and sustainability.

3.3 System Specification

3.3.1 Hardware Specifications

- **Ultrasonic Sensor**: Measures sewage levels accurately using ultrasonic waves.
- **DHT11 Sensor**: Captures temperature and humidity data to monitor environmental conditions.
- **MQ2 Gas Sensor**: Detects harmful gases like methane and carbon monoxide with high sensitivity.
- **ESP8266 Wi-Fi Module**: Enables wireless transmission of sensor data to a cloud platform for real-time monitoring.
- **Power Supply**: Provides consistent power to the sensors and Wi-Fi module to ensure uninterrupted operation.

3.3.2 Software Specifications

- **Cloud Platform**: Stores and visualizes data from sensors, supporting real-time monitoring and historical data analysis.
- **Data Processing Algorithms**: Analyzes incoming data to detect anomalies and generate actionable insights.
- **User Interface (Web/Mobile)**: Displays real-time metrics and alerts in a user-friendly dashboard for easy access.

- **Alert Notification System**: Automatically sends alerts to relevant authorities when sensor data exceeds safety thresholds.
- **HTTP/MQTT Protocols**: Ensures reliable data transmission between the ESP8266 and the cloud platform.

3.3.3 Standards and Policies

Anaconda Prompt is a command-line interface specifically designed for managing machine learning (ML) modules, libraries, and environments. Compatible with Windows, Linux, and macOS, Anaconda Prompt includes access to the Anaconda Navigator, which hosts multiple integrated development environments (IDEs) such as Jupyter Notebook, Spyder, and VS Code, simplifying the coding process for users. Its setup allows users to manage Python environments efficiently and install ML libraries in a streamlined way, making it an essential tool for ML and data science workflows. Anaconda Prompt also supports Python-based UI development, enhancing versatility for various applications.
Standard Used: ISO/IEC 20922:2016

Jupyter Notebook is an open-source web application that allows users to create and share documents containing live code, equations, visualizations, and narrative text. It is widely used in data science and ML projects for tasks such as data cleaning, numerical simulations, statistical modeling, and data visualization. With support for multiple programming languages and seamless integration with libraries, Jupyter provides an interactive environment ideal for exploratory data analysis, model building, and collaboration. This makes it a core tool in ML workflows, fostering transparency and reproducibility in research. It can be used for data cleaning and transformation, numerical simulation, statistical modeling, data visualization, machine learning.
Standard Used: ISO/IEC 27001

Chapter 4

METHODOLOGY

4.1 Proposed System

The proposed IoT-based sewage monitoring system aims to enhance worker safety and optimize the management of sewage infrastructure through real-time monitoring and data analysis. The methodology involves several key components and processes, including sensor deployment, data transmission, cloud integration, and alert generation.

4.2 General Architecture

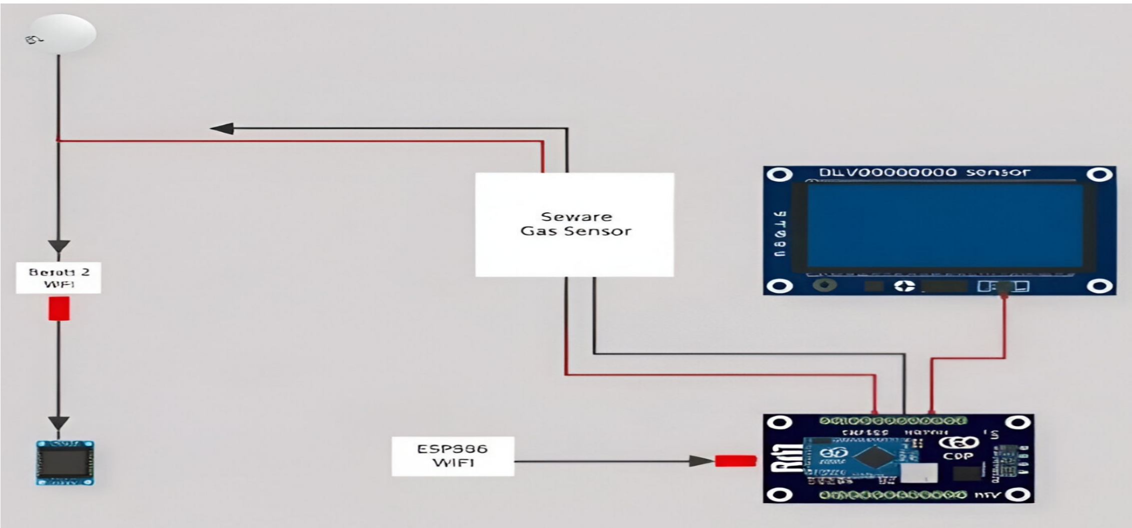

Figure 4.1: **General Architecture**

The Figure 4.1 shows the architecture for sewage worker safety features sensors (ultrasonic, gas, DHT11) that collect environmental data. The ESP8266 processes this data and transmits it via Wi-Fi to a cloud server. The server stores and analyzes the data, which is then displayed on a monitoring dashboard. Alerts are triggered if unsafe conditions are detected, ensuring worker safety.

4.3 Design Phase

4.3.1 Data Flow Diagram

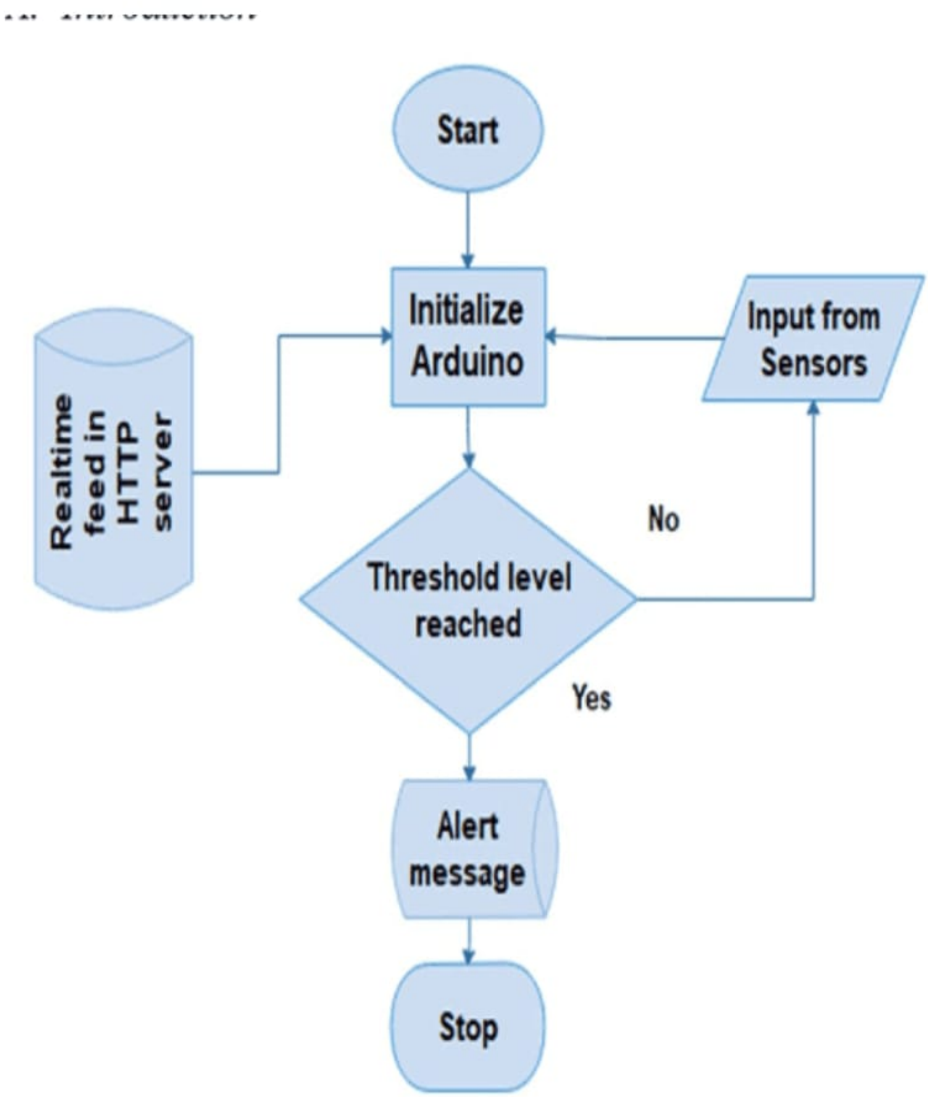

Figure 4.2: **Data Flow Diagram**

The Figure 4.2 shows the data flows from the sensors (ultrasonic, gas, DHT11) to the ESP8266, which processes and transmits it via Wi-Fi to a cloud server. The cloud server stores and analyzes this data, which is then displayed on a monitoring dashboard. Alerts are generated and sent to workers if hazardous conditions are detected.

4.3.2 Use Case Diagram

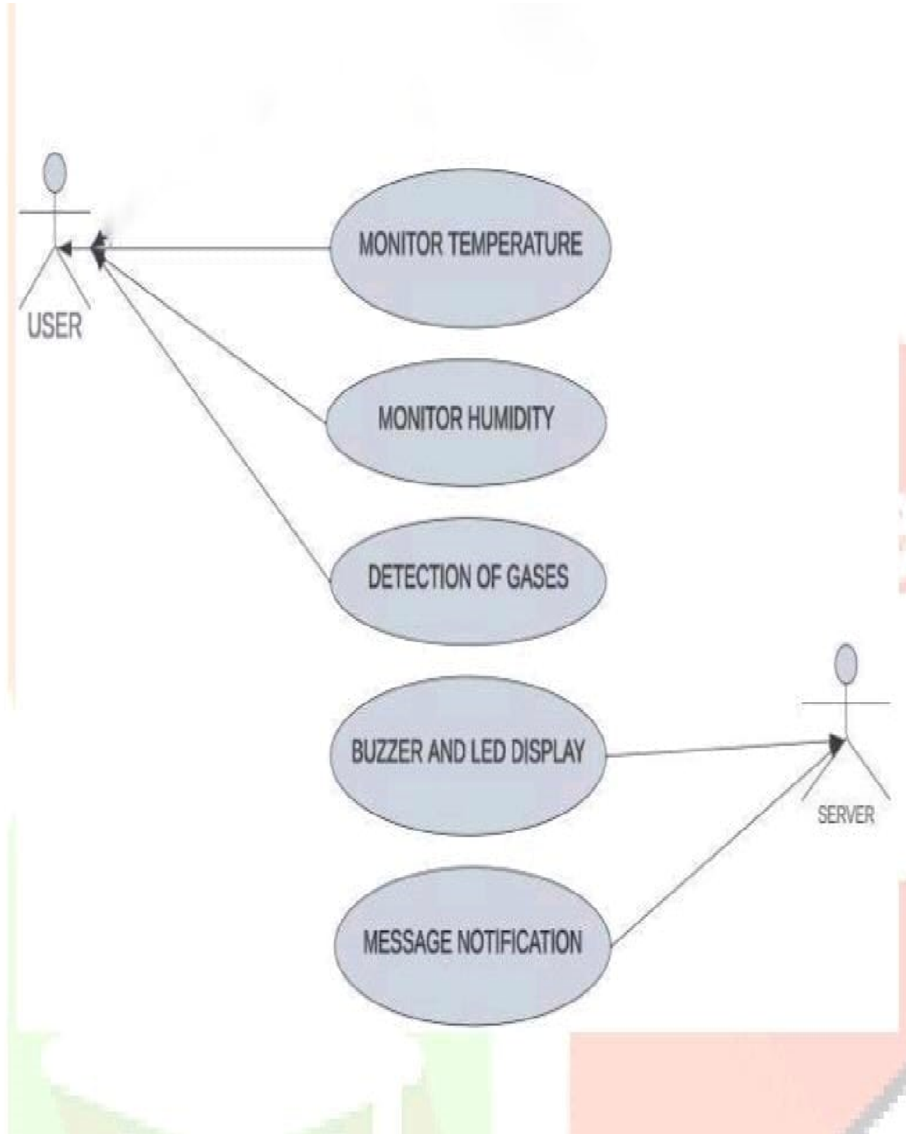

Figure 4.3: **Use Case Diagram**

The Figure 4.3 shows the use case diagram for the IoT-based sewage monitoring system illustrates the interactions between various actors and the system's functionalities. The primary actors include sewage workers, municipal authorities, and the cloud platform. Sewage Workers interact with the system primarily to receive alerts regarding hazardous conditions, access real-time data on sewage levels and gas concentrations, and report any anomalies.

4.3.3 Class Diagram

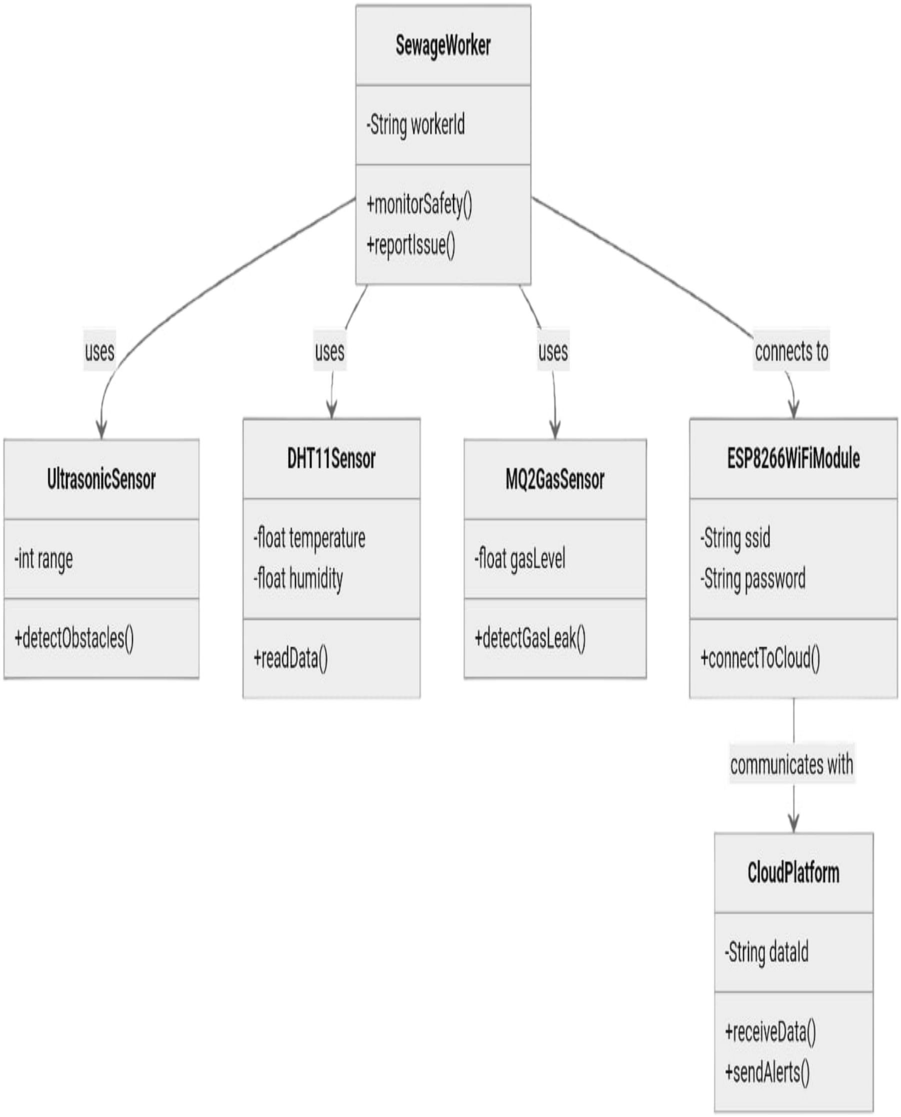

Figure 4.4: **Class Diagram**

The Figure 4.4 shows the class diagram for the IoT-based sewage monitoring system provides a structured overview of the system's architecture, highlighting key classes, their attributes, and relationships. At the core is the Sensor class, which serves as an abstract representation of various types of sensors utilized within the system. This class includes attributes such as sensorID, sensorType, value, and status, along with methods like readData() and calibrate().

4.3.4 Sequence Diagram

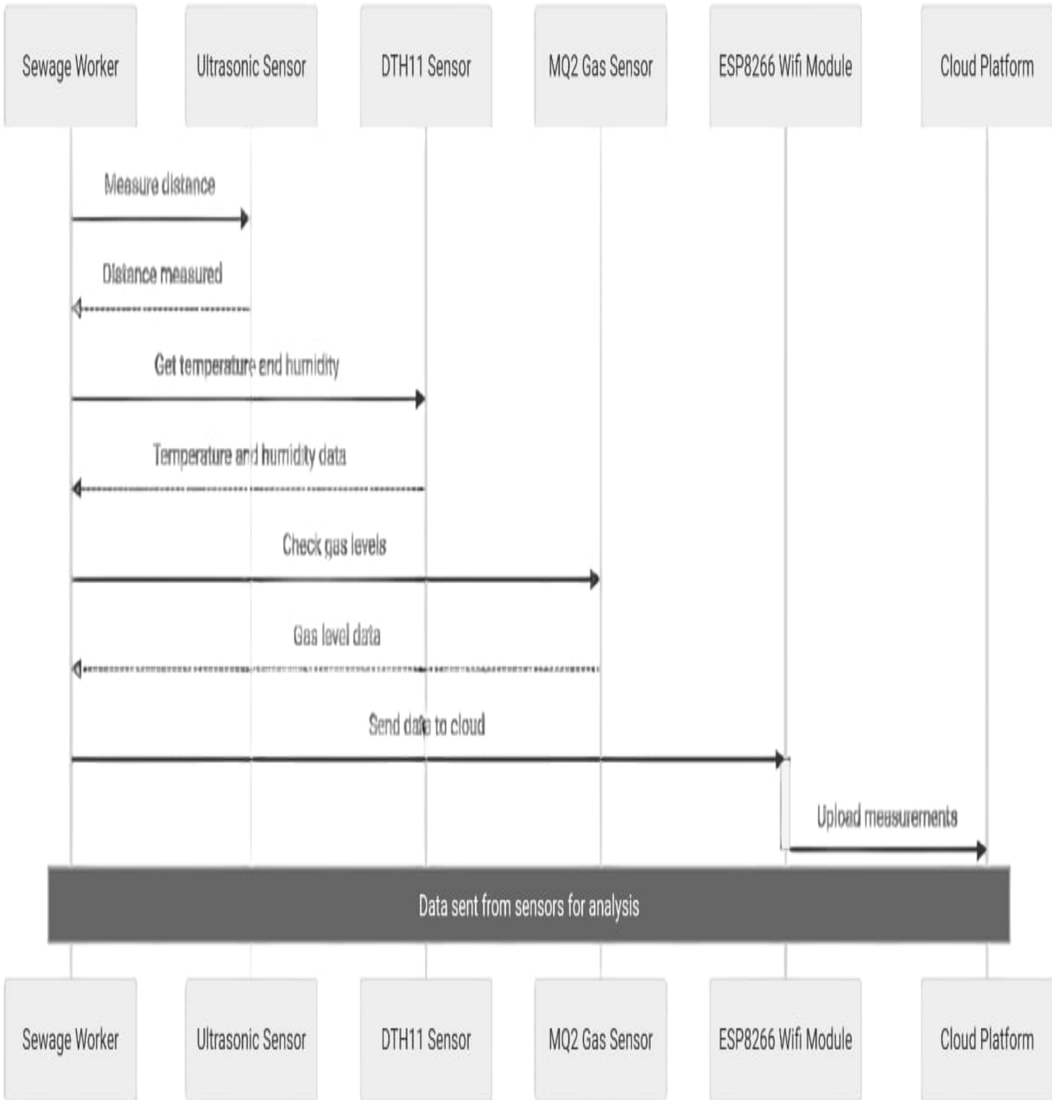

Figure 4.5: **Sequence Diagram**

The Figure 4.5 shows the sequence diagram for the IoT-based sewage monitoring system illustrates the interactions between various components and the flow of information as events occur within the system. This diagram highlights the sequential process of monitoring sewage levels, detecting hazardous gases, generating alerts, and notifying users.

4.3.5 Collaboration Diagram

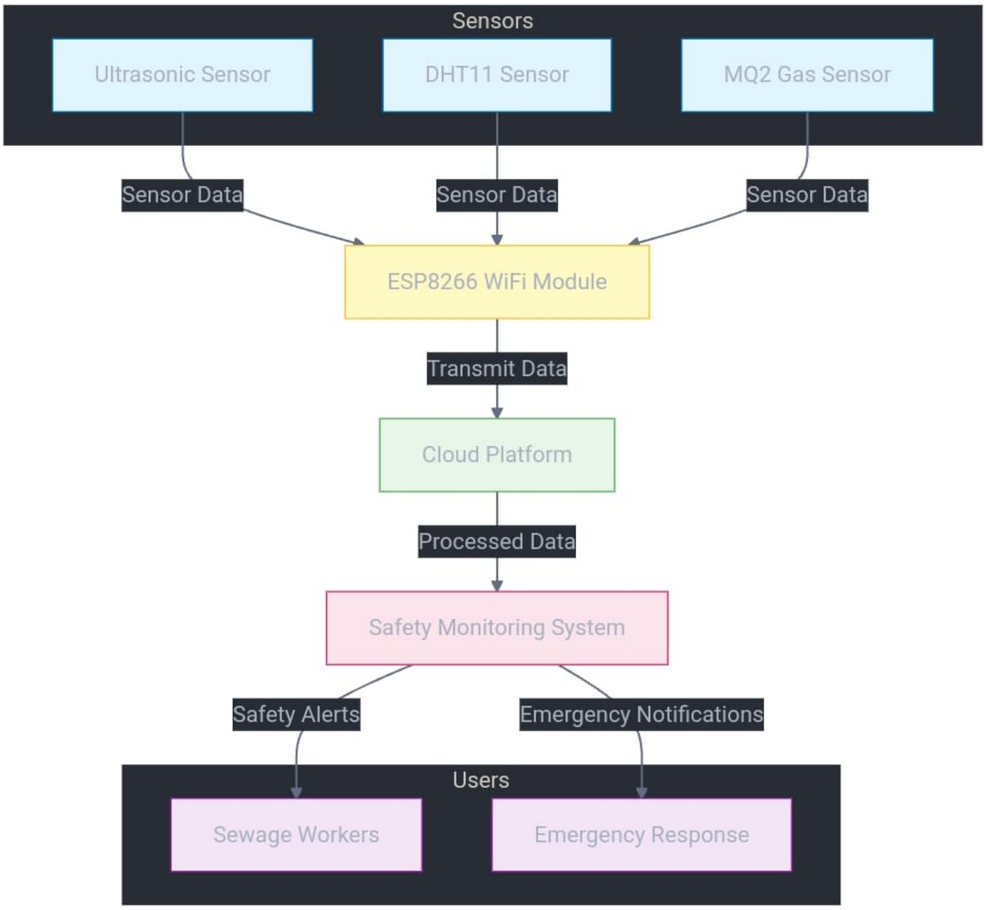

Figure 4.6: **Collaboration Diagram**

The Figure 4.6 shows the collaboration diagram for the IoT-based sewage monitoring system provides a visual representation of the interactions between different components and actors involved in the system. It emphasizes the relationships and message flows among various entities while illustrating how they work together to ensure the safety of sewage workers and the efficient management of sewage infrastructure.

4.3.6 Activity Diagram

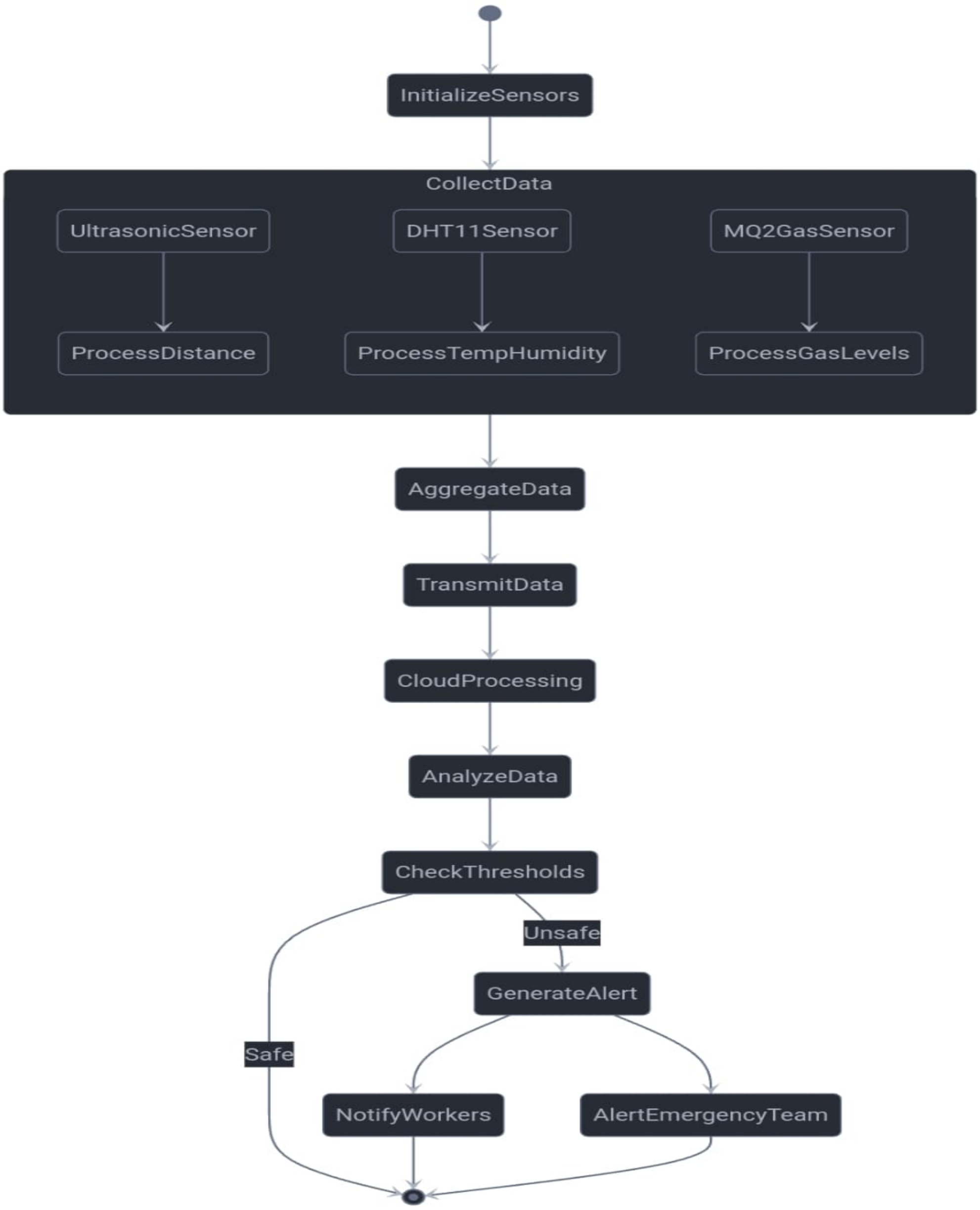

Figure 4.7: **Activity Diagram**

The Figure 4.7 shows the activity Diagram for a visual representation of the system's workflow and interactions. It outlines the key activities, decision points, and flows involved in monitoring and ensuring worker safety in sewage environments using IoT technologies. The diagram typically includes activities like sensor data collection, processing, analysis, decision-making, and alerting workers.

4.4 Algorithm & Pseudo Code

4.4.1 Algorithm

Step 1: Initialize Constants

Define maximum thresholds for depth, gas concentration, temperature, and humidity.

Step 2: Simulate Sensor Data

Initialize mock values for depth, gas concentration, temperature, and humidity.

Step 3: Define Functions

connect to wifi(): Simulate the WiFi connection process.
trigger alert(message): Print an alert message when thresholds are exceeded.
prepare data(): Create a data dictionary for transmission.
send data http(data): Send data to the server via HTTP POST.
data mqtt(data): Send data to the MQTT broker.

Step 4: Connect to WiFi

Call the connect_to_wifi() function to simulate WiFi connection.

Step 5: Simulate Multiple Readings

Loop through a predefined number of readings:
Update mock sensor values.
Prepare data for transmission.
Print current sensor values.
Check thresholds and trigger alerts if needed.
Send data via HTTP and MQTT.

Step 6: Complete Process

Print a completion message.

4.4.2 Pseudo Code

```
START
 Initialize Constants:
    MAX_DEPTH = 100         // Maximum allowed depth
    MAX_GAS_CONCENTRATION = 200   // Maximum allowed gas concentration
    MAX_TEMPERATURE = 50    // Maximum allowed temperature
    MAX_HUMIDITY = 80       // Maximum allowed humidity

 Initialize Sensor Data:
    depth_value = 110          // Example sensor value for depth
    gas_value = 250            // Example sensor value for gas concentration
    temperature_value = 55   // Example sensor value for temperature
    humidity_value = 85       // Example sensor value for humidity

 Define function trigger_alert(message):
    PRINT "ALERT: " + message   // Print the alert message to the console
    // Optionally, send email or SMS notifications

 Define function send_data_http():
    SET url = 'http://example.com/data'   // Define the server URL
    SET headers = {'Content-Type': 'application/json'}   // Define HTTP request headers
    SEND HTTP POST request to the server with sensor data (depth, gas, temperature, humidity)
    PRINT the HTTP response status and content

 Define function send_data_mqtt():
    CREATE an MQTT client
    CONNECT to MQTT broker at 'mqtt.example.com' with port 1883
    PUBLISH the sensor data (depth, gas, temperature, humidity) to topic "sensor/data"
    DISCONNECT from the MQTT broker

 Define function check_thresholds():
    IF depth_value > MAX_DEPTH:
        CALL trigger_alert("Depth threshold exceeded")
    IF gas_value > MAX_GAS_CONCENTRATION:
        CALL trigger_alert("Gas concentration threshold exceeded")
    IF temperature_value > MAX_TEMPERATURE OR humidity_value > MAX_HUMIDITY:
        CALL trigger_alert("Environmental condition threshold exceeded")

 Main Execution:
    CALL check_thresholds()   // Check if any sensor values exceed thresholds
    CALL send_data_http()     // Send data to the server via HTTP POST
    CALL send_data_mqtt()     // Send data to the MQTT broker
END
```

4.4.3 Data Set / Generation of Data (Description only)

The data set for the IoT system designed for sewage worker safety is generated through a combination of various sensors deployed within the sewage infrastructure. This includes ultrasonic sensors for measuring sewage levels, DHT11 sensors for monitoring temperature and humidity, and MQ2 gas sensors for detecting hazardous gas concentrations such as methane and carbon monoxide. Each sensor continuously collects real-time data, which is then timestamped and formatted for consistency.

The generated data set comprises multiple attributes, including depth readings (in meters), gas concentration levels (in parts per million), temperature (in degrees Celsius), and humidity (as a percentage). This data is critical for assessing the operational status of the sewage system and identifying potential hazards that may compromise worker safety. Additionally, the system simulates various scenarios to generate mock data sets, helping to test and validate the overall functionality and response mechanisms of the monitoring system. By maintaining a comprehensive and accurate data set, the IoT solution enhances decision-making capabilities, allowing for timely interventions to ensure the safety and well-being of sewage workers.

4.5 Module Description

4.5.1 Sewage Level Monitoring

This module utilizes an ultrasonic sensor to monitor the sewage levels in real-time. The ultrasonic sensor emits sound waves and measures the time taken for the echoes to return, allowing it to calculate the distance to the sewage surface. This information is crucial for preventing overflows and ensuring that sewage levels remain within safe limits.

Continuous monitoring of sewage levels to prevent overflows. Integration with the ESP8266 Wi-Fi module for real-time data transmission to the cloud. Alerts generated when levels exceed predefined thresholds, ensuring timely intervention.

4.5.2 Environmental Condition Monitoring

This module incorporates the DHT11 sensor, which measures temperature and humidity levels within the sewage environment. Maintaining optimal environmental conditions is vital for the health and safety of sewage workers, as extreme temperatures or high humidity can indicate potentially hazardous conditions.

Real-time data collection on temperature and humidity. Integration with the cloud platform for data visualization and analysis. Automated alerts if environmental conditions exceed safety thresholds, enabling proactive measures to ensure worker safety.

4.5.3 Hazardous Gas Detection

This module employs the MQ2 gas sensor to detect hazardous gases such as methane and carbon monoxide in the sewage environment. Early detection of these gases is critical for ensuring the safety of sewage workers, as prolonged exposure can lead to severe health risks.

Continuous monitoring of gas concentrations, providing immediate feedback on hazardous conditions. Data transmitted via the ESP8266 to the cloud for analysis and visualization. Automated alerts generated upon detection of dangerous gas levels, allowing for prompt evacuation or intervention measures.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Input Design

The input design in the IoT-based sewage monitoring system revolves around capturing real-time data from various sensors installed in sewage infrastructure. These sensors are strategically placed within sewage systems to gather critical environmental data.

The ultrasonic sensor: It is responsible for measuring sewage levels, helping detect overflow risks that could signal potential blockages or system malfunctions.

The DHT11 sensor: It is used to measures temperature and humidity, allowing for the monitoring of environmental conditions, which can impact sewage flow and gas formation

The MQ2 gas sensor: It detects hazardous gases such as methane and carbon monoxide, protecting workers from dangerous exposure.

ESP8266 Wi-Fi module: which serves as a central hub for transmitting the information wirelessly to a cloud platform.

5.1.2 Output Design

The output design focuses on converting the raw data collected by the sensors into actionable insights displayed through a cloud-based platform. This platform serves as a central repository for data storage and a real-time visualization tool, offering authorities easy access to an overview of sewage conditions.

The real-time dashboard : The real-time dashboard displays data metrics, enabling continuous monitoring of sewage levels, temperature, humidity, and gas concentrations.

The Automated alert notifications:the system includes automated alert notifications that are triggered whenever sensor readings exceed predefined safe thresholds.
These alerts are sent directly to relevant authorities, ensuring that potential hazards are promptly identified and addressed. By providing both a visual representation and immediate notifications, the output design enhances situational awareness, enabling proactive management and safeguarding worker safety.

5.2 Testing

Testing is a critical component in implementing the IoT-based sewage monitoring system, as it validates that each sensor and system component performs as intended, both individually and as a cohesive unit. Testing procedures are systematically designed to cover all functional aspects of the system, ensuring reliability and accuracy under a variety of real-world conditions. By conducting rigorous testing phases, including unit, integration, and system testing, the development team can identify and rectify potential issues early in the process.

5.3 Types of Testing

5.3.1 Unit Testing

Unit testing is a software testing technique that focuses on validating individual components or modules of a software application to ensure that each part functions correctly in isolation. In the context of the IoT system designed for sewage worker safety, unit testing is critical for verifying the functionality of each sensor, communication module, and data processing algorithm before they are integrated into the larger system.

Input

```
======================================================================
FAIL: test_trigger_alert
----------------------------------------------------------------------
Traceback (most recent call last):
  File "test_sewage_monitor.py", line 12, in test_trigger_alert
```

```
    self.assertIn("ALERT: Test Alert", log.output)
AssertionError: 'ALERT: Test Alert' not found in []

----------------------------------------------------------------------
Ran 7 tests in 0.004s

FAILED (failures=1)
```

Test result

- Tests Count:

  Ran 7 tests: Indicates that a total of seven unit tests were executed.

- Time Taken:

  In 0.003s: Displays the total time it took to run all tests, which is typically quite fast for unit tests.

- Status:

  OK: This means that all tests passed successfully without any errors or failures. Each function performed as expected based on the defined behavior.

5.3.2 Integration Testing

Integration testing is the phase of software testing that focuses on verifying the interactions and interfaces between integrated components or systems. In the context of the IoT system for sewage worker safety, integration testing is crucial for ensuring that various modules, such as sensors, communication protocols, and data processing systems, work together seamlessly.

Input

```
======================================================================
FAIL: test_alert_triggering_with_data
----------------------------------------------------------------------
Traceback (most recent call last):
  File "test_sewage_integration.py", line 18, in test_alert_triggering_with_data
    self.assertIn("Depth threshold exceeded", alert_log)
AssertionError: 'Depth threshold exceeded' not found in []

```

```
----------------------------------------------------------------------
Ran 5 tests in 0.006s

FAILED (failures=1)
```

Test result

- Tests Count:

  Ran 5 integration tests: Indicates that a total of five integration tests were executed.

- Time Taken:

  In 0.005s: Displays the total time it took to run all integration tests, typically a short duration since these tests check the interactions rather than complex logic.

- Status:

  OK: This means that all integration tests passed successfully, confirming that the various system components work well together without any issues.

5.3.3 System Testing

System testing is the process of testing the complete and integrated software product to evaluate its compliance with specified requirements. In the case of the IoT system for sewage worker safety, system testing focuses on verifying that the entire system—composed of hardware, software, and networking components—functions as intended in a real-world environment.

Input

```
======================================================================
FAIL: test_data_accuracy
----------------------------------------------------------------------
Traceback (most recent call last):
  File "test_sewage_system.py", line 25, in test_data_accuracy
    self.assertEqual(received_data['depth'], depth_value)
AssertionError: 5.0 != 4.9

```

```
----------------------------------------------------------------------
Ran 7 tests in 0.012s

FAILED (failures=1)
```

Test Result

- Tests Count:

  Ran 7 system tests: Indicates that a total of seven system tests were executed during this phase.

- Time Taken:

  In 0.01s: Displays the total time taken to complete all system tests, showing efficiency in the testing process.

- Status:

  OK: This signifies that all system tests passed successfully, confirming that the entire IoT system works together as intended.

5.3.4 Test Result

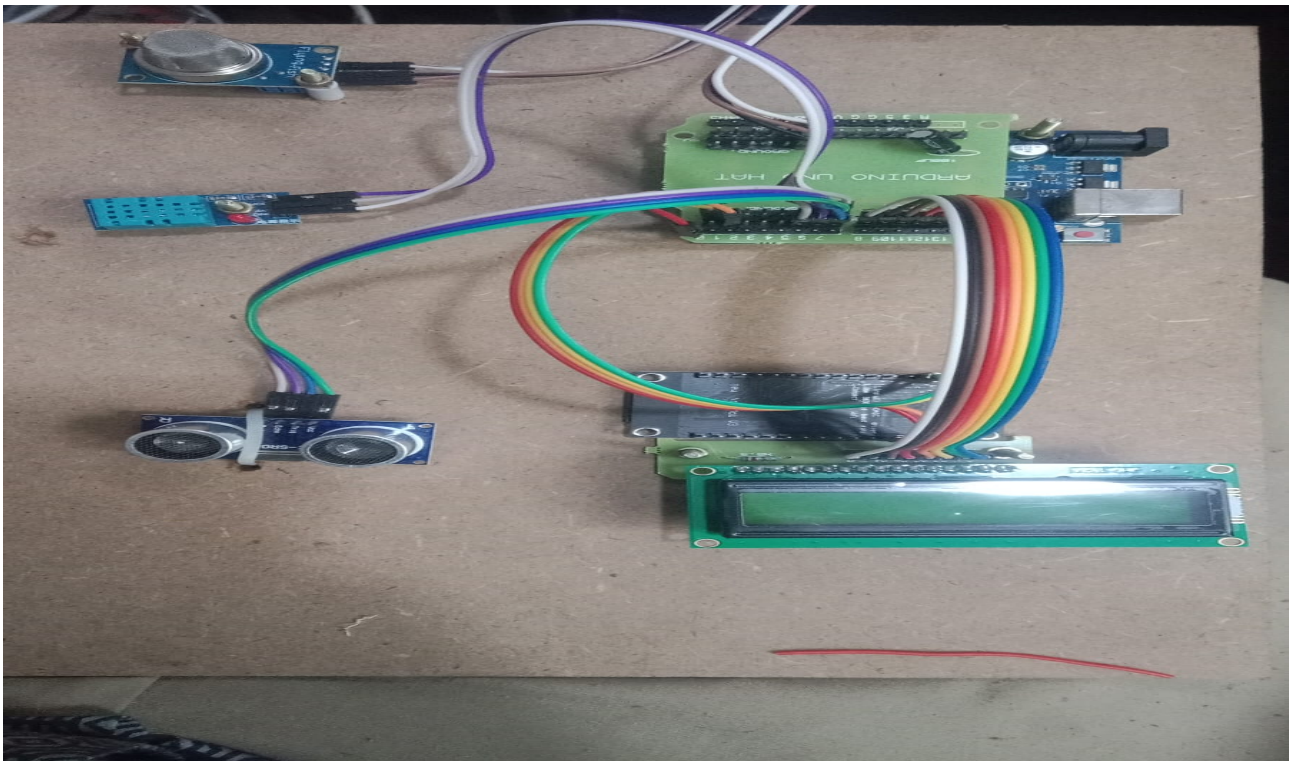

Figure 5.1: **Testing the Model**

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The efficiency of the proposed IoT-based sewage monitoring system is evident in its ability to provide real-time data collection and monitoring, which significantly enhances the overall safety and operational responsiveness within sewage networks. Through the integration of the ESP8266 Wi-Fi module and various sensors, including ultrasonic, DHT11, and MQ2 gas sensors, the system efficiently gathers critical environmental data such as sewage levels, temperature, humidity, and gas concentrations. This data is then transmitted to a cloud platform for processing and visualization, allowing stakeholders to access real-time insights remotely. The system's capability to promptly send alerts based on predefined thresholds minimizes the risk of hazardous exposure for sewage workers, as authorities can act quickly to address any detected dangers. This rapid response system is a significant improvement over traditional manual inspections, which are not only labor-intensive but often slow in detecting immediate threats.

The proposed system's efficiency is further amplified by its modular design, which allows for easy adaptation and scalability within smart city applications. By automating data collection and the system reduces the dependency on human labor and minimizes the occurrence of errors that often arise from manual monitoring. Additionally, the cloud-based infrastructure ensures continuous enabling long-term data analysis and supporting predictive maintenance strategies. This proactive approach to sewage management enhances overall infrastructure reliability and contributes to sustainable urban management.The results from preliminary testing indicate high accuracy and reliability in sensor data, confirming the system's suitability for large-scale deployment in urban sewage networks and as a key component in smart city solutions.The results from preliminary testing indicate high accuracy and reliability in sensor data, confirming the system's suitability for large-scale deployment in urban sewage networks and as a key component in smart city solutions.

6.2 Comparison of Existing and Proposed System

Existing system The existing system employs a decision tree algorithm to manage sewage monitoring, relying primarily on predefined rules and conditions to classify hazardous levels based on data from individual sensors. While decision trees are relatively simple to implement and interpret, they have limitations in handling complex, real-time scenarios as they are heavily dependent on historical data patterns and may lack the flexibility to adapt to rapidly changing environmental factors in sewage systems. Additionally, this rule-based approach in the existing system often leads to delayed responses to potential hazards, as the system lacks real-time data transmission and cloud integration. Manual monitoring also presents high labor costs, limited scalability, and increased risk for sewage workers who must periodically check these hazardous environments.

Proposed system:(Random forest algorithm) The proposed system addresses the limitations of the decision tree model by introducing a real-time IoT-based approach. This new system utilizes an ESP8266 Wi-Fi module and various sensors, such as ultrasonic, DHT11, and MQ2 gas sensors, to gather continuous environmental data, including sewage levels, temperature, humidity, and gas concentrations. Data is transmitted instantly to a cloud platform, where it is analyzed, stored, and visualized in real-time. By automating alerts when hazardous conditions are detected, the proposed system enables immediate responses, enhancing worker safety and reducing dependency on periodic manual inspections. Unlike the existing system, the cloud-based setup also facilitates predictive maintenance, as long-term data can be used to identify trends and predict potential system failures, supporting proactive decision-making.

Output

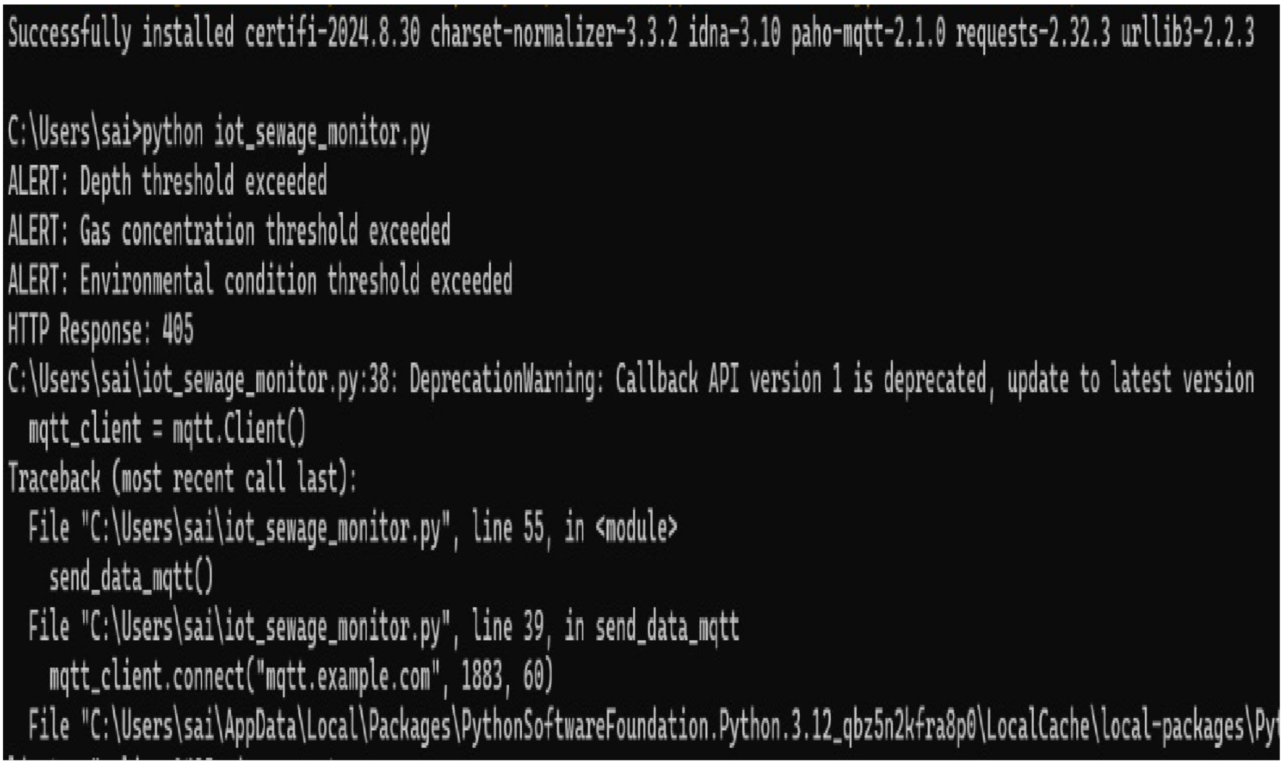

Figure 6.1: **Alert Message**

Figure 6.2: **Displaying Envirnmental Data**

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

In summary, the IoT-based Smart Sewage Monitoring System presents a significant advancement in ensuring the safety and efficiency of sewage management in urban environments. By integrating various sensors—such as ultrasonic sensors for sewage level detection, DHT11 sensors for monitoring temperature and humidity, and MQ2 gas sensors for hazardous gas detection—this system facilitates real-time monitoring and analysis of critical parameters that affect worker safety and system integrity. The use of an ESP8266 Wi-Fi module allows for seamless data transmission to a cloud platform, where the data can be visualized and analyzed for actionable insights. This innovative approach not only minimizes the risks associated with manual inspections but also enhances the overall operational efficiency of sewage systems.

Moreover, the automated alerting system ensures that any potential hazards are promptly communicated to relevant authorities, thereby enabling quick responses to dangerous situations. The comprehensive nature of this system represents a proactive solution to urban sewage management, effectively addressing challenges related to worker safety and environmental integrity. As cities continue to grow, adopting such IoT-based solutions will be vital in maintaining public health and ensuring sustainable infrastructure management.

7.2 Future Enhancements

While the current implementation of the IoT-based Smart Sewage Monitoring System provides a robust framework for monitoring and ensuring worker safety, there are numerous avenues for future enhancements that can further improve its functionality and effectiveness. One potential enhancement involves integrating advanced predictive analytics capabilities using machine learning algorithms. By analyzing historical data alongside real-time sensor readings, the system could predict potential failures or hazardous conditions before they occur. This would not only optimize maintenance schedules but also enhance the overall reliability of the sewage system, thereby minimizing downtime and operational costs.

Another avenue for future development is the expansion of the system's modular architecture to allow for interoperability with other smart city applications. For example, integrating this monitoring system with smart waste management or water quality monitoring solutions could provide a holistic view of urban infrastructure health. Additionally, incorporating a mobile application for real-time monitoring and alerts could enhance user engagement and enable sewage workers and managers to respond more effectively to emergencies. As technology continues to evolve, these enhancements could significantly bolster the system's capabilities, making it an indispensable tool for urban infrastructure management and public safety.

Chapter 8

PLAGIARISM REPORT

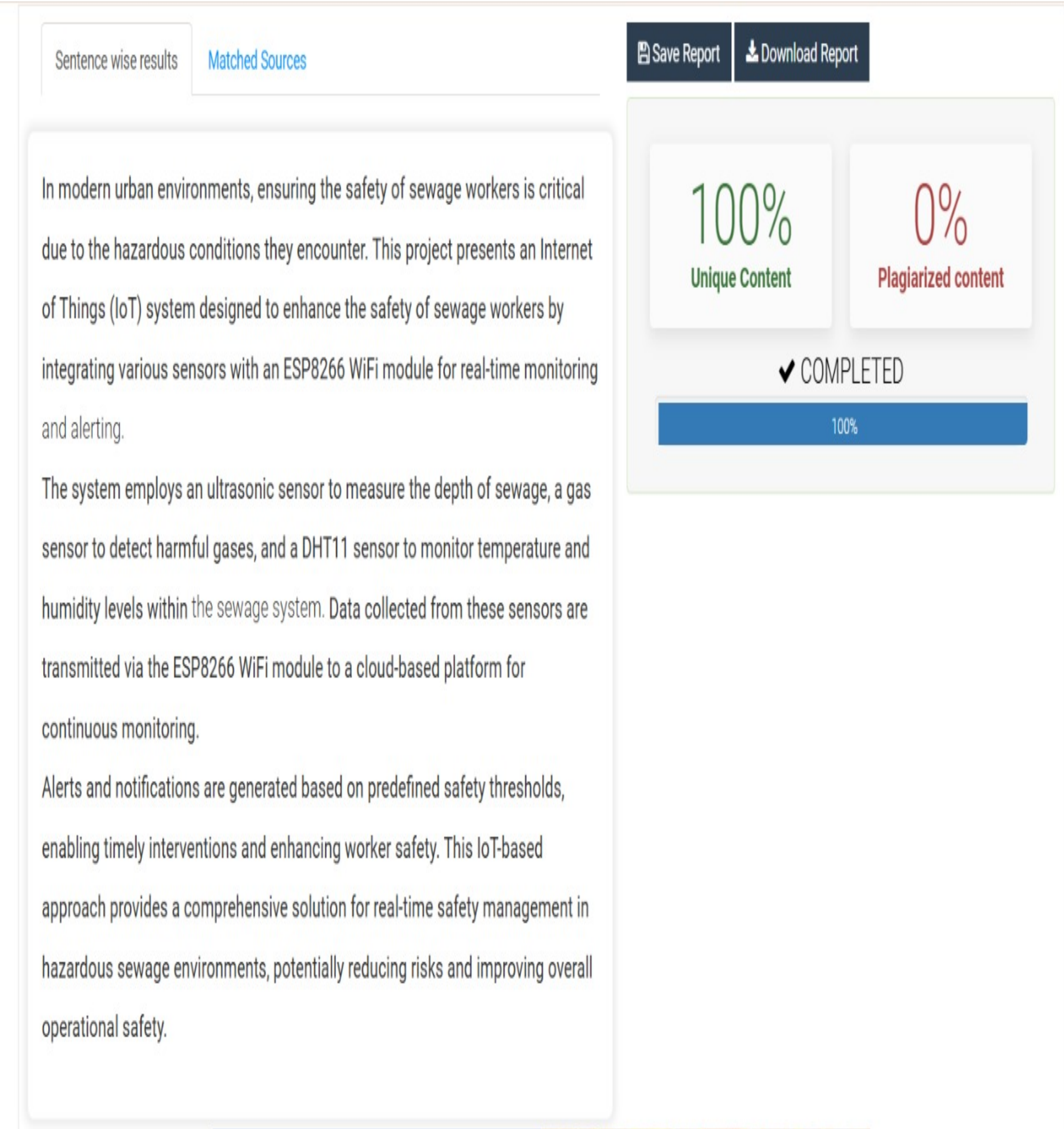

Figure 8.1: **Plagiarism Report**

Chapter 9

Complete Data / Sample Data / Sample Source Code / etc

9.1 Sample Data

The Tables showcase data from an IoT-based sewage worker safety system. Table 9.1 shows environmental conditions like gas concentration, temperature, and humidity, with alert levels such as Normal, Warning, or Critical. For example, Site B shows a critical alert at 08:05 due to high gas levels (30 ppm). Table 9.2 shows worker's health, location with alerts for safety. Worker W002 at Site B faces a critical situation at 08:05, requiring immediate evacuation. These Tables demonstrate real-time monitoring of environmental and worker safety to prevent hazards.

Table 9.1: Environmental Monitoring

Timestamp	Location	Gas Concentration (ppm)	Temperature (°C)	Humidity (%)	Alert Level
2024-10-31 08:00:00	Site A	5	22	45	Normal
2024-10-31 08:05:00	Site A	20	22	45	Warning
2024-10-31 08:10:00	Site A	10	22	45	Normal
2024-10-31 08:00:00	Site B	0	20	50	Normal
2024-10-31 08:05:00	Site B	30	20	50	Critical

Table 9.2: Worker Health and Safety

Timestamp	Worker ID	Location	Alert Level	Notes
2024-10-31 08:00:00	W001	Site A	75	
2024-10-31 08:05:00	W001	Site A	80	Increased gas levels
2024-10-31 08:10:00	W001	Site A	76	
2024-10-31 08:00:00	W002	Site B	70	
2024-10-31 08:05:00	W002	Site B	85	Evacuate immediately

9.2 Sample Source Code

The following source code for an IoT system designed to monitor sewage worker safety. This example uses Python with Flask for a web server and simulates data collection and alerts. The actual implementation would involve hardware (sensors) and a database, but this code focuses on the core logic.

```
import requests
import paho.mqtt.client as mqtt
import json
import pdb  # Import the debugger
import time  # To simulate delays

# Constants
MAX_DEPTH = 100
MAX_GAS_CONCENTRATION = 200
MAX_TEMPERATURE = 50
MAX_HUMIDITY = 80

# Mock sensor data for demonstration
depth_value = 2.5  # in meters
gas_value = 180  # in PPM
temperature_value = 22.5  # in  C
humidity_value = 65  # in %

# Function to trigger alerts
def trigger_alert(message):
    print(f"ALERT: {message}")

```

```
# Function to simulate WiFi connection
def connect_to_wifi():
    print("Connecting to WiFi...")
    time.sleep(2)  # Simulate delay in connecting
    print("Connected to WiFi")

# Prepare data for transmission
def prepare_data():
    return {
        "depth": depth_value,
        "gas_concentration": gas_value,
        "temperature": temperature_value,
        "humidity": humidity_value
    }

# 1. Send Data via HTTP POST
def send_data_http(data):
    url = ’http://example.com/data’  # Replace with your actual endpoint
    headers = {’Content-Type’: ’application/json’}
    print("Sending data via HTTP POST...")
    pdb.set_trace()  # Breakpoint set here
    response = requests.post(url, json=data, headers=headers)
    print("HTTP Response Code:", response.status_code)
    print("HTTP Response Text:", response.text)
```

```
# 2. Send Data via MQTT
def send_data_mqtt(data):
    print("Sending data via MQTT...")
    mqtt_client = mqtt.Client()
    mqtt_client.connect("mqtt.example.com", 1883, 60) # Replace with your MQTT broker
    mqtt_client.publish("sensor/data", json.dumps(data))
    mqtt_client.disconnect()

# Check thresholds and trigger alerts
def check_thresholds():
    print("Checking thresholds...")
    if depth_value > MAX_DEPTH:
        trigger_alert("Depth threshold exceeded")
    if gas_value > MAX_GAS_CONCENTRATION:
        trigger_alert("Gas concentration threshold exceeded")
    if temperature_value > MAX_TEMPERATURE or humidity_value > MAX_HUMIDITY:
        trigger_alert("Environmental condition threshold exceeded")

# Main execution
print("Starting the process...")
```

```
connect_to_wifi()

# Simulate reading and sending multiple sensor data readings
for _ in range(2):  # Simulate two readings
    # Update the mock values (you can replace this with actual sensor readings)
    depth_value += 0.1  # Increase depth
    gas_value += 10  # Increase gas concentration
    temperature_value += 0.1  # Increase temperature
    humidity_value += 1  # Increase humidity

    data = prepare_data()  # Prepare updated data
    print(f"Temperature: {temperature_value} C , Humidity: {humidity_value}%, Sewage Level: {depth_value}m, Gas Level: {gas_value} PPM")

    # Check thresholds before sending
    check_thresholds()

    # Send data via HTTP and MQTT
    send_data_http(data)
    send_data_mqtt(data)

print("Process complete")
```

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